# MultiModal LLM Key Components

| Component | **Modality Encoders** | **Input Projector** | **LLM Backbone** | **Output Projector** | **Modality Generator** |
| --- | --- | --- | --- | --- | --- |
| Primary Function | Transform raw data from various modalities (image, audio, 3D, etc.) into numerical feature representations (embeddings). | Align encoded features from different modalities into a common space, typically compatible with the LLM backbone's input. Typically using MLPs, cross attentions, Q-Formers etc. | Serves as the central reasoning and language processing engine, integrating aligned multimodal information. Usually pretrained LLMs, say GPT, Gemini, LLaMA etc. | Maps LLM outputs (e.g., signal tokens for generation) into features suitable for modality-specific generators. Typically MLPs or Transformers. | Produces outputs in non-textual modalities (e.g., images, audio) based on input from the Output Projector. This can be StableDiffusion, AudioLDM etc |

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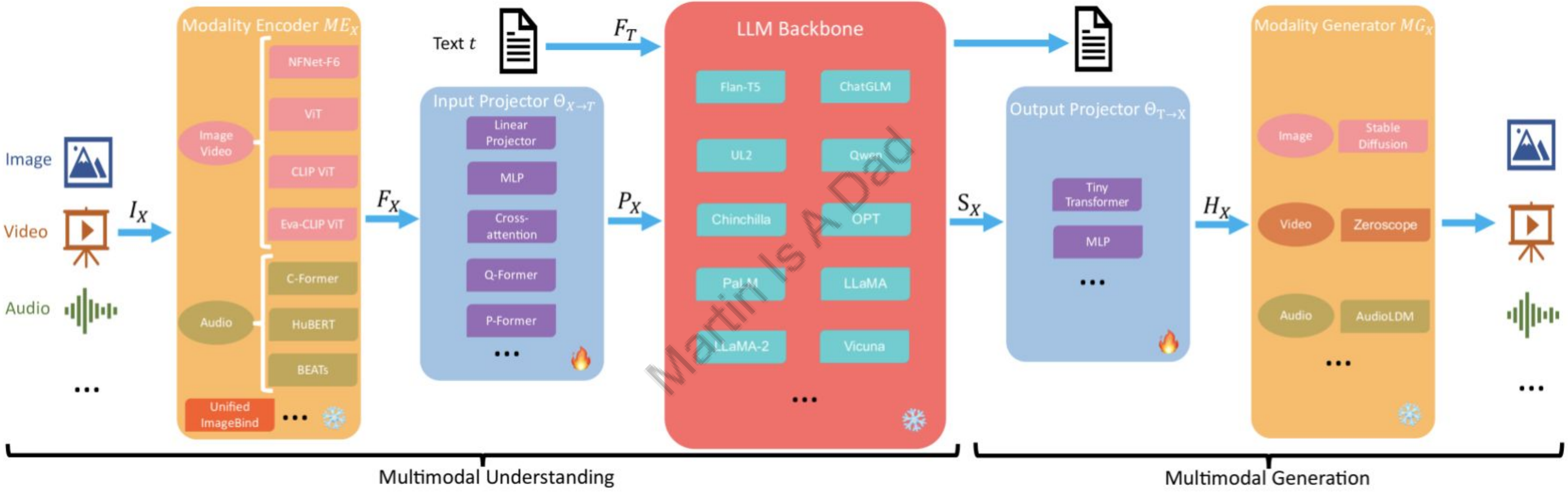
# MultiModal LLM Key Components



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*From: [arxiv 2401.13601v1] MM-LLMs: Recent Advances in MultiModal Large Language Models*

# LLaVA: GPT-assisted Visual Instruction Generation

**Context type 1: Captions**
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.

**Context type 2: Boxes**
person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

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**Response type 1: conversation**
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

**Response type 2: detailed description**
The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

**Response type 3: complex reasoning**
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

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*From: [arxiv 2304.08485] Visual Instruction Tuning*

# LLaVA: Architecture

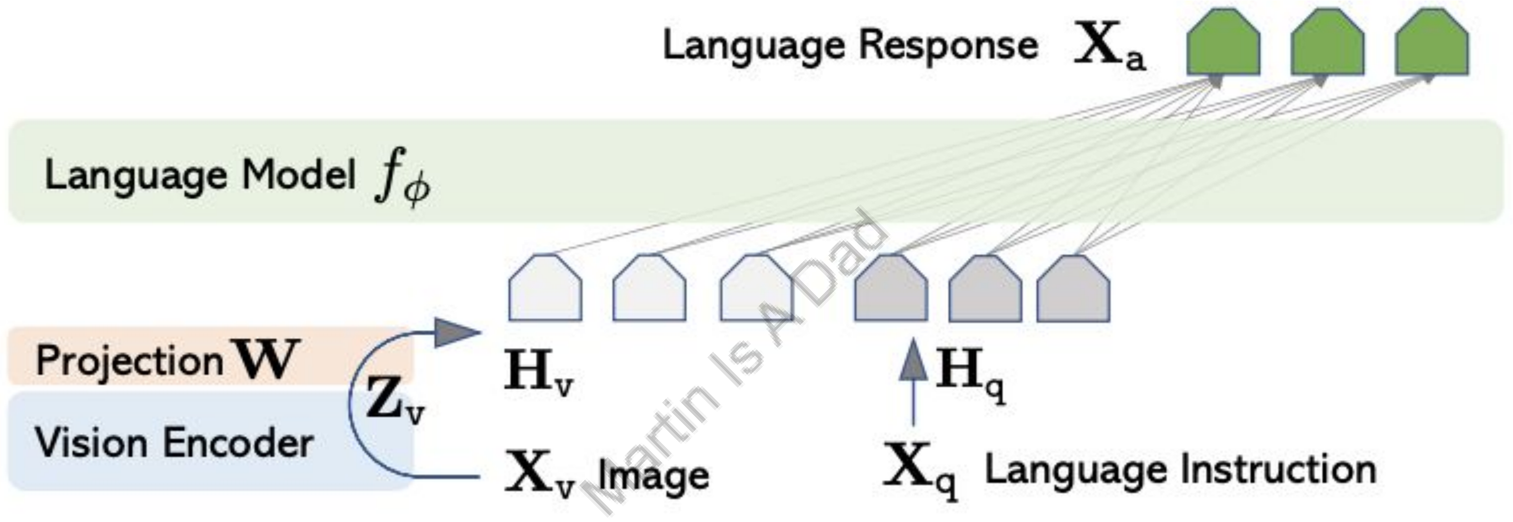


Figure 1: LLaVA network architecture.

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*From: [arxiv 2304.08485] Visual Instruction Tuning*

# BLIP-2: Overview

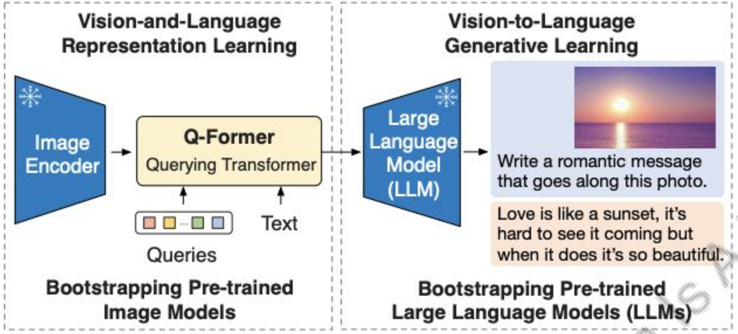


*Figure 1.* Overview of BLIP-2's framework. We pre-train a lightweight Querying Transformer following a two-stage strategy to bridge the modality gap. The first stage bootstraps vision-language representation learning from a frozen image encoder. The second stage bootstraps vision-to-language generative learning from a frozen LLM, which enables zero-shot instructed image-to-text generation (see Figure 4 for more examples).

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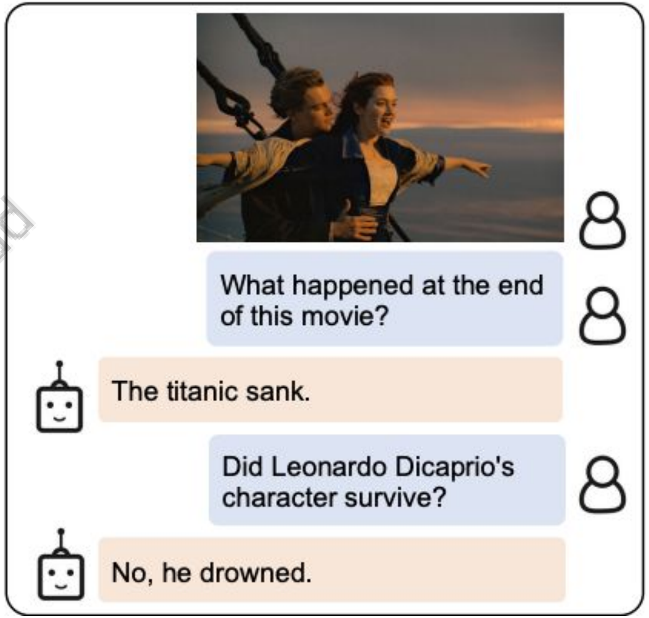
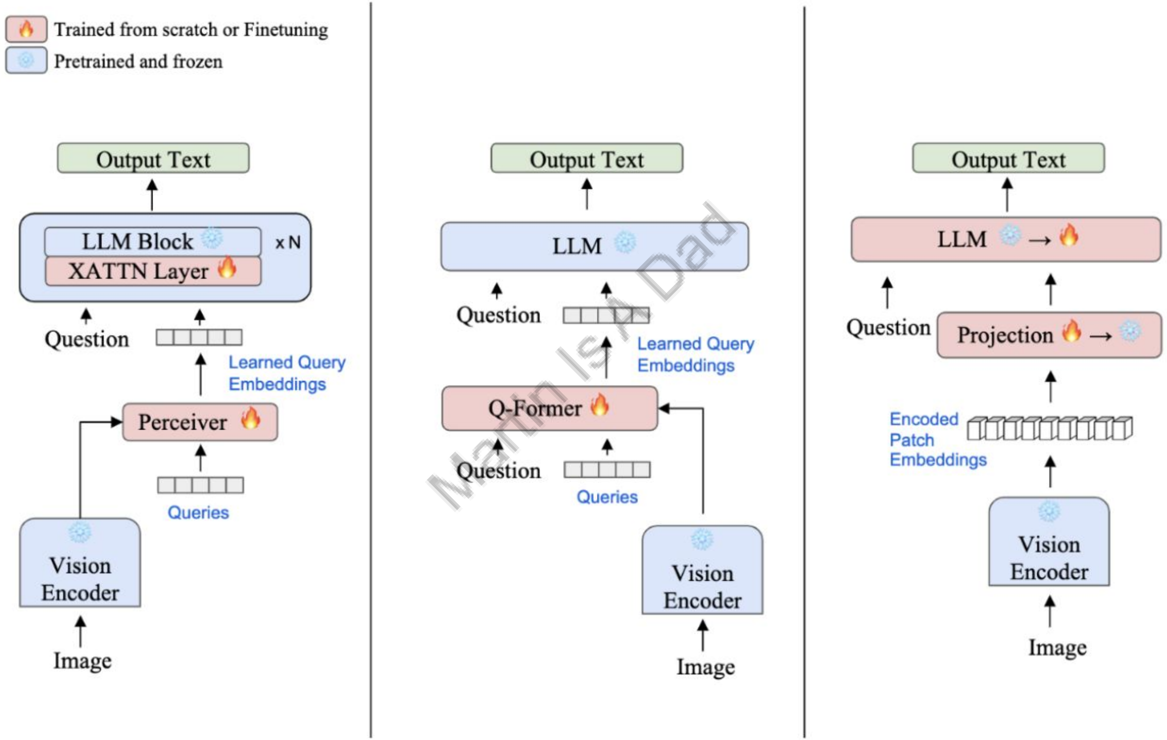



*From: [arxiv 2301.12597] BLIP-2*

# MLLM Training Comparison

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Flamingo

BLIP-2 / InstructBLIP

LLaVA

*From Sayna Ebrahimi*

# Current LLM's Fake Reasoning

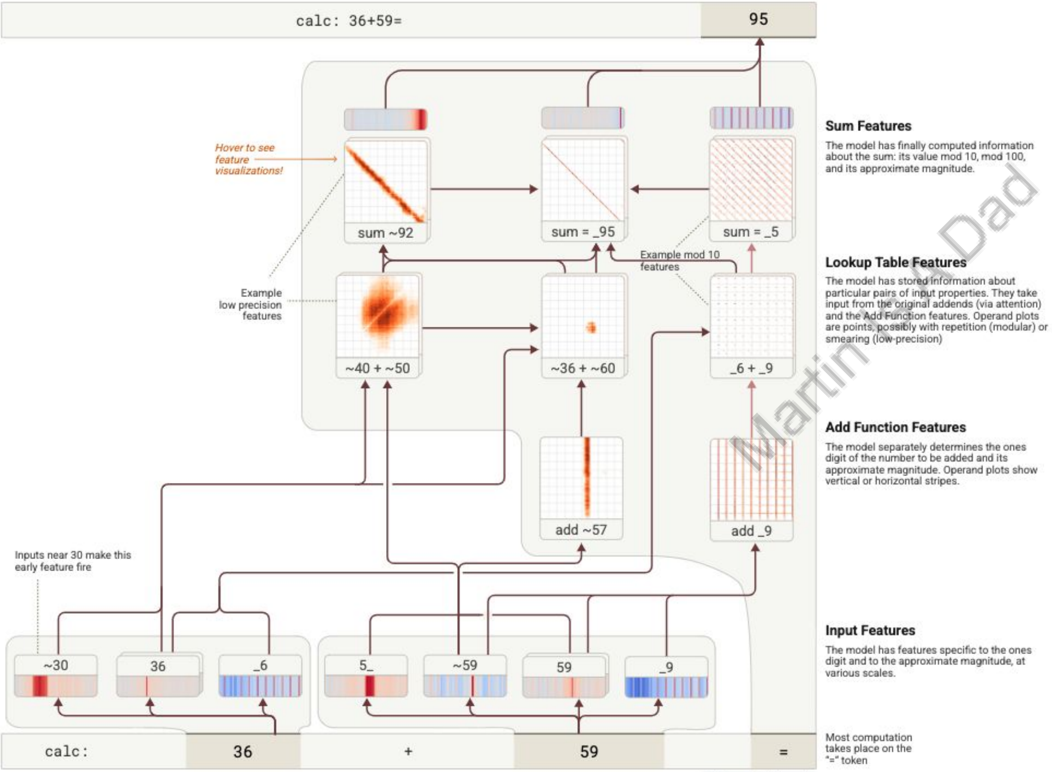



**Figure 26**: A simplified attribution graph of Haiku adding two-digit numbers. Features of the inputs feed into separable processing pathways.

View detailed graph

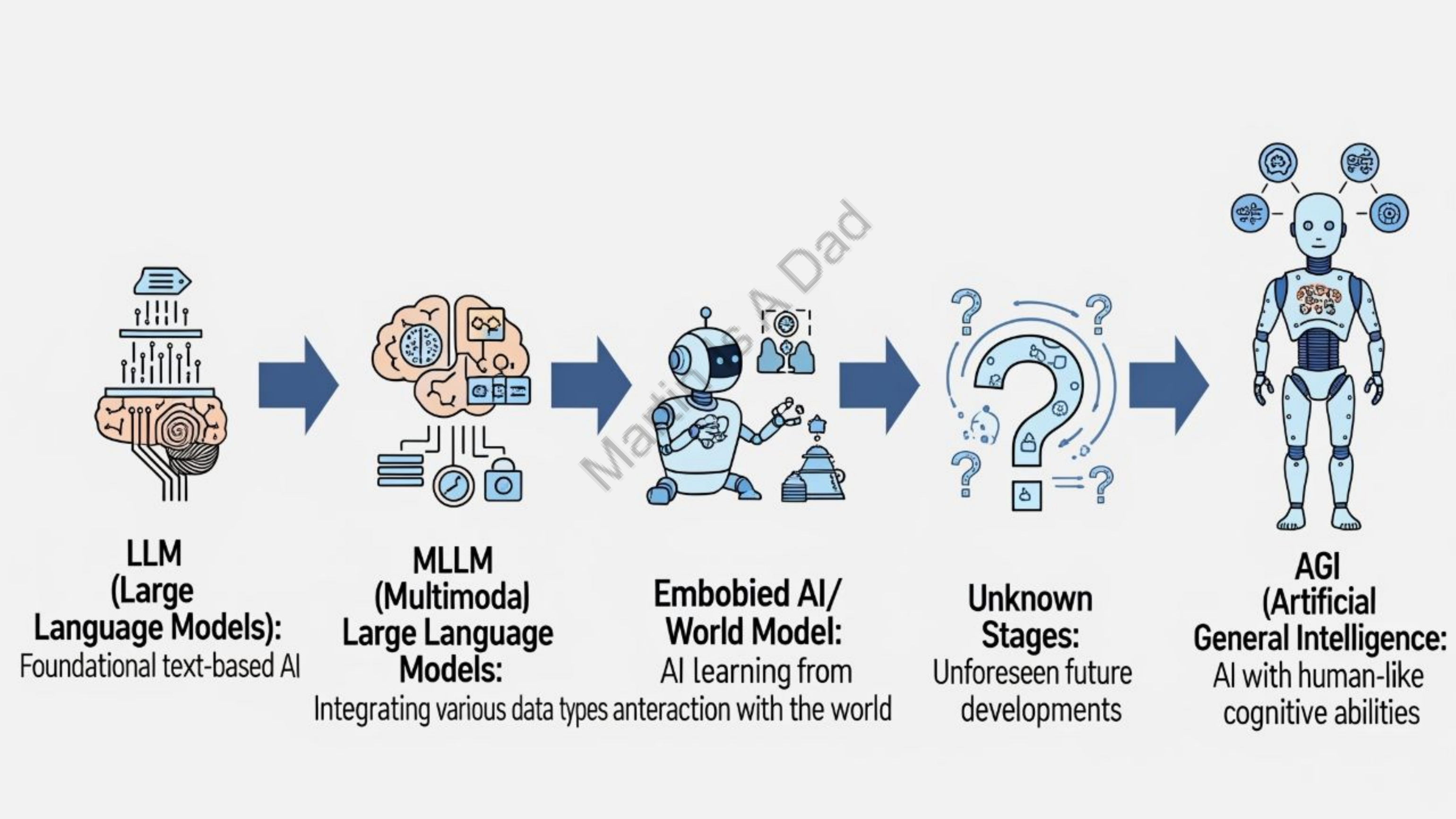
```
Human: Answer in one word. What is 36+59?

Assistant: 95

Human: Briefly, how did you get that?

Assistant: I added the ones (6+9=15), carried the 1, then added the tens
(3+5+1=9), resulting in 95.
```

# Future



LLM
(Large
Language Models):
Foundational text-based AI
MLLM
(Multimoda)
Large Language
Models:
Integrating various data types
Embobied AI/
World Model:
AI learning from
anteraction with the world
Unknown
Stages:
Unforeseen future
developments
AGI
(Artificial
General Intelligence:
AI with human-like
cognitive abilities
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